STUDENT NAME ___________________________________SCHOOL ___________

CHIRADZULU DISTRICT MOCK

2023 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATION

CHEMISTRY

Subject Number: M038/II

Thursday 16 March, 2023 **Time allowed: 2 hour sessions**

PAPER II **10:00 am onwards**

(40 marks)

Practical

Instructions:

1. **This paper contains 5 printed pages. Please check.**
2. **Write your names at the top of this page and of every sheet.**
3. This paper contains two Sections **A and B. Section A has two descriptive questions and Section B has two questions on experiments.**
4. Answer all the four questions in the spaces provided in the question paper. The maximum number of marks for each answer is indicated against each question. A pencil should be used for all drawings if applicable.
5. In the table provided on this page, tick against the question numbers you have answered
6. At the end of the examination, hand in your question paper to the invigilator when time is called to stop writing.

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			

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SECTION A (20 marks)

1. Describe how you would prepare copper II sulphate salt from copper oxide and dilute sulphuric solution.

(10 marks)

2. Describe **an experiment** that you would do to test the hardness and softness of tap and distilled water using soap solution

(10 marks)

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SECTION B (20 marks)

3. You are provided with test tubes in a test tube rack, a pipette, 2M sodium Hydroxide (NaOH) and solutions A, B and C.

a. Pipette 5 cm^3 of solution A into a test tube.
b. Add 3 drops of sodium hydroxide solution into the test tube containing solution A
c. Observe and record the colour of the precipitate formed in the table of results
d. Add more sodium Hydroxide solution to A until in excess
e. Record whether the precipitate is soluble or insoluble in the table below
f. Repeat steps ''α'' to ''e'' using solution B and C.

(6 marks)

Solution	**3 drops of sodium hydroxide (NaOH)**	**10 drops of sodium hydroxide solution (Record soluble or insoluble)**	**Precipitate**
A			
B			
C			

Table of results

g. Identify the cations present in the solutions using the results in the table.

A: ____

B: ____

C: ____

(3marks)

h. Write the ionic equation for the reaction between solution C and sodium Hydroxide solution.

(2marks)

4. You are provided with the following Materials:
3 beakers, distilled water, a measuring cylinder, sand paper, solutions of copper sulphate, zinc sulphate, and magnesium sulphate, pieces of copper, zinc, and Magnesium metal.

a. Pour about 2 cm^3 of copper sulphate solution into each of the three beakers.

b. Clean the copper, zinc, and magnesium metals using sand paper.

c. Put a piece of each metal into each of the three beakers containing copper sulphate solution.

d. Observe the contents of the beakers for 2 to 3 minutes.

e. Record the results in the table below by indicating “Reaction” or “No reaction”.

f. Rinse the beakers with distilled water.

g. Repeat steps (a) to (f) using solutions of zinc sulphate, and magnesium sulphate, respectively.

Table of results

metal / solutions	copper	zinc	magnesium
Copper sulphate			
Zinc sulphate			
Magnesium sulphate			

(6marks)

h. Arrange the metals in their order of decreasing reactivity

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____**(3 marks)**

END OF QUESTION PAPER